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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2021

Time: 3 hours

Code: 100203 ***Subject: Chemistry*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven questions in brief: **[2x7=14]**

(a) State de Broglie's principle.

(b) What is an orbital?

(c) What are the shapes of BF_3 and ClF_3 molecules?

(d) Out of NH_3, PH_3, AsH_2 and SbH_2, which possesses largest bond angle? Give reason.

(e) Out of Cr^{2+} and Cr^{3+}, which one is stable in aqueous solution?

(f) Cu^+ is colourless but Cu^{2+} is coloured. Why?

(g) What is selection rule?

(h) A gas expands against vacuum. What is the work done on it?

(i) What is the condition for a reaction to be in equilibrium?

(j) The presence of CO_2 in boiler feed water should be avoided. Why?

Q.2 Solve both questions : **[9+5=14]**

(a) A water sample had the following constituents per litre:

$CaCO_3$= 160mg
$MgHCO_3$= 150mg
$CaSO_4$= 136mg
$MgSO_4$= 120mg
$NaCl$= 10mg

Calculate the quantity of temporary and permanent hardness in the water sample. Calculate the quantity of lime (78% purity) and soda (92% purity) required for softening of 25 million litre of above water sample.

(b) In an experiment to determine the hardness of a sample of water, 50 mL of N/50Na_2CO_3 solution was added to 200 mL of water sample. After complete precipitation of insoluble carbonate, the unreacted Na_2CO_3 was titrated against N/50H_2SO_4 solution, when 20 mL of acid was required. Calculate the hardness and comment on the nature of hardness so determined.

Q.3 Solve both questions : **[10+4=14]**

(a) Write short notes on the following:

(i) Photoelectric effect (ii) Heisenberg's uncertainty principle

(b) How many photons of light having a wavelength of 3000 Å are necessary to provide 1 J of energy? ($h = 6{\cdot}626\times10^{-34}$J−s)

Q.4 Solve both questions : **[6+8=14]**

(a) Explain the behaviour of CO as ligand with different metal ions using molecular orbital theory.

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(b) Draw the MO energy-level diagram for NO molecule. Using this diagram, calculate and explain bond order and magnetic behaviour of (i) NO, (ii) NO^+ and (iii) NO^-.

Q.5 Solve all parts : **[6+4+4=14]**

(a) Which type of electronic transition(s) is/are observed in UV-visible spectrum of aniline in the range 200 nm to 900 nm? Justify your answer with suitable figure.

(b) The internuclear distance of NaCl is $2{\cdot}36\times10^{-10}$m. Calculate the reduced mass and moment of inertia of NaCl. (Atomic mass of Cl = 35×10^{-3}kg mol^{-1} and Na = 23×10^{-3}kg mol^{-1})

(c) Calculate the force constant for CO, if it absorbs at $2{\cdot}143\times10^{5}$m^{-1}. (Atomic mass of C = 12×10^{-3}kg mol^{-1} and O = 16×10^{-3}kg mol^{-1})

Q.6 Solve both questions : **[6+8=14]**

(a) How many types of ^{1}H NMR signals are expected for (i) $CH_3COOCH_2CH_3$ and (ii) CH_3CHCl_2? Mention the relative intensity ratio for the signal(s) observed for (i) and (ii).

(b) 0-6 mol of NH_3 at 25°C occupies a volume of 3 dm^3. Calculate the pressure using van der Waals equation (a = 0.417N m^{4}mol^{-2} and $b = 0.037\times10^{-3}$m^{3}mol^{-1}). Compare the above result with the pressure calculated using ideal gas equation.

Q.7 Solve both questions : **[7+7=14]**

(a) 7 mol of a monatomic ideal gas are compressed reversibly and adiabatically. The initial volume is 15 dm^3 and the final volume is 9 dm^3. The initial temperature is 27 °C. (i) What would be the final temperature in this process? (ii) Calculate w, q and ΔU for the process. Given, C_v= 20·91J K^{-1}mol^{-1} and γ= 1·4.

(b) Write a cell representation (in proper cell representation) whose cell reaction is $AgCl \rightarrow Ag^+ + Cl^-$, using the following standard electrode potentials at 298 K:

$E_{AgCl/Ag,\,Cl}{}^{o}$= 0·22V

$E_{Ag^+/Ag}{}^{o}$= 0·80V

Calculate E^o of the cell. Calculate solubility product (or solubility constant) of AgCl and its solubility at 298 K.

Q.8 Solve both questions : **[7+7=14]**

(a) Write notes on the following:

(i) Optical isomerism of lactic acid (ii) Optical isomerism of tartaric acid

(b) Differentiate between the following:

(i) Enantiomers and diastereomers (ii) Racemic mixture and meso-compounds

Q.9 Solve both questions : **[6+4+4=14]**

(a) Explain the following:

(i) Trichloroacetic acid is stronger acid than acetic acid.

(ii) The amino group in aniline is o- and p-directing but nitro group is m-directing.

(b) Arrange the following carbocations in order of increasing stability with suitable reasons:

$C_6H_5CH_2^+$

$(CH_3)_3C^+$

$(C_6H_5)_2CH^+$

$CH_3CH_2^+$

(c) Write short notes on the following:

(i) Markownikoff's rule (ii) Kharasch's rule

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